

GAT Quantitative Ability

1) Average of five numbers is 20. If each number is multiplied by 2, what will be the new average?

- A. 30
- B. 40
- C. 50
- D. 60

The correct answer is **B**

Explanation: The new average = Initial average * 2 = $20 * 2 = 40$

2) A group consists of two male, two female and three children. The average age of the male is 67 years, that of the female is 35 years, and that of the children is six years. What is the average age of the group?

- A. 30.71
- B. 31.71
- C. 28.71
- D. 35.45

Answer: B

Explanation: Average = $(67*2+35*2+6*3)/7$
 $= (134+70+18) / 7$
 $= 222 / 7$
 $= 31.71$

3) The captain of a cricket team of 11 members is 26 years old, and the wicket-keeper is three years older than the captain. If the ages of captain and wicketkeeper are excluded, the average age of the remaining players of the team is one year less than the average age of the whole team. What is the average age of the team?

- A. 19
- B. 49
- C. 17
- D. 23

Answer: D

Explanation:

Let the average age of the whole team by x years.

$$11x - (26 + 29) = 9(x - 1)$$

$$11x - 55 = 9x - 9$$

$$11x - 9x = -9 + 55$$

$$2x = 46$$

$$x = 23 \text{ Years.}$$

4) A museum has an average of 510 visitors on Sunday and 240 on other days. Find the average number of visitors per day in a month of 30 days beginning with a Sunday.

A. 285

B. 275

C. 237

D. 245

Answer: A

Explanation: Since, the month begins with a Sunday, so there will be 5 Sundays and 25 other days in this month.

So, the average no. of visitor per day

$$\frac{510 \cdot 5 + 240 \cdot 25}{30} = 285$$

5) If the number of quantities in group A is 10 and in group B is 8, and their individual average is 24 and 16 respectively, find the combined average of the two groups.

A. 20.44

B. 18.22

C. 16.22

D. 18.66

Answer: A

Explanation:

The combined average of the two groups is =

$$\frac{\text{No. of quantities in Gp. A} \times \text{Average of Gp. A} + \text{No. of quantities in Gp. B} \times \text{Average of Gp. B}}{\text{No. of quantities in Gp. A} + \text{Number of quantities in Gp. B}} = (10 \cdot 24 + 8 \cdot 16) / (10 + 8)$$

$$\begin{aligned}
 &= (240+128)/18 \\
 &= 368/18 \\
 &= 20.44
 \end{aligned}$$

6) Find the average of cubes of first 7 consecutive natural numbers.

- A. 112
- B. 110
- C. 116
- D. 113

Answer: A

Explanation:

The average of cubes of first n consecutive natural numbers is given by;

$$= n(n+1)^2/4$$

Here, n=7

$$= 7(7+1)^2/4$$

$$= 7*64/4$$

$$= 448/4$$

$$= 112$$

7) The average weight of a group of 5 boys is 26. If we replace a boy of weight 25 in the group with another boy so that new average increases by 3, find the weight of the new body.

- A. 56
- B. 34
- C. 40
- D. 33

Answer: C

Explanation:

The weight of new person = Weight of the removed person + No. of persons * Increase in average.

$$= 25+5*3$$

$$= 25+15$$

$$= 40$$

4) 7X2 is a three digit number and X is the missing digit. If the number is divisible by 6, the missing digit is

- A. 4
- B. 3
- C. 2
- D. 5

The Correct answer is (B)

Answer with explanation: The given number is divisible by 6 so it would be divisible by 2 and 3. As the last digit is 2 whatever be the value of X, it would be divisible by 2.

Now, $7+X+2 = 9 + X$, must be divisible by 3.

$\therefore X = 3$ makes the number divisible by 3, so 3 is the required digit.

7) If $(2^p + 1)$ is a prime number, which one of the following digits could be the value of p?

- A. 3
- B. 4
- C. 5
- D. 6

The Correct answer is (B)

Answer with explanation: In such questions, each alternative should be tried.

So, if $P = 3$, we will get; $8 + 1 = 9$

If $P = 4$, we will get; $16 + 1 = 17$

If $P = 5$, we will get; $32 + 1 = 33$

If $P = 6$, we will get; $64 + 1 = 65$

Out of the four results, only 17 is the prime number. So, the required value of the P is 4.

8) A boy was asked to multiply a number by 25 but by mistake he multiplied by 45 and the answer was 200 more than the correct answer. What was the number?

- A. 7
- B. 8

- A. 4
- B. 2

C. $\sqrt{64}$

D. 8

Answer: D

Solution: In case of same type of root values multiplication, the number itself will be the answer i.e., the number is 8.

Hence, the answer is 8.

11) $(1-1/2)(1-1/3)(1-1/4) \dots (1-1/x) = ?$

A. x

B. x-1

C. $1/(x-1)$

D. $1/x$

Answer: D

Explanation: This type of question can solve by a sort trick that is the first term numerator, and the last term denominator will be the answer.

i.e., $1-1/2 = 1/2$, $1-1/3 = 2/3$, $1-1/4 = 3/4$ and so on

$(1/2) * (2/3) * (3/4) \dots ((x-1)/x) = (1) (1/x)$

The first term denominator cancelled by its next term numerator till the end.

In the end, we get $1/x$.

12) Find the number which is completely divisible by 45.

A. 306990

B. 331145

C. 256987

D. 181660

Answer: A

Explanation: A number, which is divisible by 3, 5, and 9, is also divisible by 45.

A number is divisible by 3 if the sum of its entire digits is divisible by 3.

A number is divisible by 9 if the sum of its entire digits is divisible by 9.

The number which ends with 0 or 5 is divisible by 5.

The number 306990 fulfils all the requirements, so the answer is 306990.

13) On a certain sum of money the compound interest Rs. 318 is earned in 2 years. If the rate of interest is 12%, what is the principal amount?

- A. Rs. 1250
- B. Rs.1300
- C. Rs. 1200
- D. Rs. 1150

The Correct answer is (A)

Explanation:

We have;

$$\text{C.I.} = \text{Rs. } 318$$

$$r = 12\%$$

$$t = 2 \text{ years}$$

$$\text{Apply formula: C.I} = P \left[\left(1 + \frac{r}{100} \right)^t - 1 \right]$$

$$318 = P \left[\left(1 + \frac{12}{100} \right)^2 - 1 \right]$$

$$318 = P \left[\frac{112}{100} * \frac{112}{100} - 1 \right]$$

$$318 = P * 0.2544$$

$$P = \frac{318}{0.2544} = \text{Rs. } 1250 \text{ (Option A)}$$

14) If the difference between Simple Interest and Compound Interest on a sum of money for 3 years is Rs. 200 and the rate of interest is 10%, what is the sum?

- A. Rs. 6451.6
- B. Rs. 6351.6
- C. Rs. 6251.6
- D. Rs. 6151.6

The Correct answer is (A)

Explanation: If the difference between C.I. and S.I. on a certain sum for 3 years at rate of interest $r\%$ is Rs. X, then the sum is given by;

$$\text{Sum} = \frac{X * (100)^3}{r^2 (300 + r)}$$

We have;

$$X = \text{Rs. } 200$$

$$r = 10\%$$

Therefore, sum is:

$$\begin{aligned} &= \frac{200 * 100 * 100 * 100}{10 * 10 * 310} \\ &= \frac{2000000}{310} = \text{Rs. } 6451.6 \text{ (Option A)} \end{aligned}$$

15) A's and B's shares in a business are in the ratio of 5:3. If A has invested Rs. 70000 for 12 months, for what period B has invested Rs. 60000?

- A. 7 months
- B. 7.4 months
- C. 8 months
- D. 8.4 months

The Correct answer is (D)

Explanation:

A's capital be $C_1 = 70000$

B's capital be $C_2 = 60000$

A's time be $T_1 = 12$ months

Let B's time be $T_2 = X$ months

A's share: B's share = 5:3

Apply formula:
$$\frac{\text{Profit of A}}{\text{Profit of B}} = \frac{C_1 * T_1}{C_2 * T_2}$$

$$\frac{5}{3} = \frac{C1 \cdot T1}{C2 \cdot T2}$$

$$\frac{5}{3} = \frac{70000 \cdot 12}{60000 \cdot X}$$

$$300000 X = 2520000$$

$$X = \frac{252}{30} = 8.4 \text{ months}$$

16) X and Y enter into a partnership for a year. X invests Rs. 6000, and Y invests Rs. 8000. After 4 months, they admit Z, who invests Rs. 9000. If Y withdraws his contribution after 6 months, how would they share a profit of Rs 1000 at the end of the year?

- A. 350, 300, 350
- B. 375, 250, 375
- C. 400, 300, 300
- D. 100, 600, 300

Answer: B

Explanation:

X's capital be C1 = 6000

Y's capital be C2 = 8000

Z's capital be C3 = 9000

X's time be T1 = 12 months

Y's time be T2 = 6 months

Z's time be T3 = 8 months

Profit = 1000

The profit will be divided in the ratio:-

(C1 * T1): (C2 * T2): (C3 * T3)

(6000*12): (8000*6): (9000*8)

i.e., 72000: 48000: 72000

Or, 72:48:72

Divide the whole equation by 24.

The ratio will be 3: 2: 3

Sum of the ratios will be 3+2+3= 8

Apply formula:

X's share = (X's ratio/ sum of all three ratios)* total profit

Hence, X's share is $(3/8) * 10000 = 375$

X's and Z's share are equal in ratio, so Z's share = 375

Y's share = $1000 - (A + B)'s \text{ share}$

= $1000 - 750 = 250$

17) The age of John 10 years ago was thrice the age of his son. After ten years, John's age will be twice that of his son. What is the ratio of their present ages?

A. 7:3

B. 3:7

C. 9:4

D. 4:9

Answer: A

Explanation:

Let the age of son 10 years ago = x years

Then age of John = 3x

According to question

After 10 years

$2(x+20) = 3x+20$

$2x + 40 = 3x + 20$

$X = 20$

The present of son and John are $x+10$ and $3x+ 10$ respectively

Son's age = $20+10 = 30$ years

John's age = $3 \times 20 + 10 = 60+10 = 70$ years

Ratio of their present age is $=70:30 = 7:3$

18) On February 5, 1998, it was Thursday. The day of the week on February 5, 1997, was

A. Wednesday

B. Monday

C. Friday

D. Sunday

The correct option is (A)

Explanation: 1997 was an ordinary year, it had 1 odd day. So, the day on February 5, 1998, would be one day beyond the day on February 5, 1997.

∴ Thursday on February 5, 1998, would be one day beyond the day on February 5, 1997, so the day on February 5, 1997, was Wednesday.

19) The day on 5th April of a year will be the same day on 5th of which month of the same year?

- A. 5th July
- B. 5th August
- C. 5th June
- D. 5th October

The Correct answer; option (A)

Explanation: April & July for all years have the same calendar. So, a day on any date of April will be the same day on the corresponding date in July.

∴ The same day will fall on 5th July of the same year.

20) A man can row a boat at a speed of 20 km/hr in still water. If the speed of the stream is 5 km/hr, in what time he can row a distance of 75 km downstream?

- A. 1.5 hours
- B. 2 hours
- C. 2.5 hours
- D. 3 hours

The correct answer is **D**

Answer with explanation: Speed of boat = 20 km/hr

Speed of stream = 5 km/hr

∴ Speed downstream = 20 + 5 = 25 km/hr

$$\text{Required Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{75}{25} = 3 \text{ hours}$$

21) A boat travels upstream from Q to P and downstream from P to Q in 3 hours. If the distance between P to Q is 4km and the speed of the stream is 1kmph, then what is the velocity of the boat in still water?

- A. 3kmph
- B. 4kmph
- C. 5kmph
- D. 7.2kmph

Answer: A

Answer with the Explanation: Let the velocity or speed of the boat in still water is x km/hr.
And the Speed of the stream = 1km/hr

So, the speed of the boat along the stream = $(x+1)$ km/hr.
The speed of the boat against the stream = $(x-1)$ km/hr.

Note: time = Distance / Speed

So, $[4 / (x+1)] + [4 / (x-1)] = 3$ hrs.

Note: go through the given options to get the answer quickly or solve the equation as follows:

Or, $[4 (x+1+x-1)] / [(x+1) (x-1)] = 3$

Or, $8x = 3(x^2-12)$

Or, $8x = 3x^2-36$

Or, $3x^2-8x-36=0$

Or, $3x^2- 9x+ x-36 = 0$

Or, $(x-3) (3x+12) = 0$

Therefore $x=3$ or, $x=-12/3$ (speed can't be -ve)

Hence, the speed or velocity of the boat in still water is 3 km/hr.

22) How many times the hands of a clock coincide in a day?

- A. 24
- B. 22
- C. 23
- D. 21

The Correct answer is (B)

Explanation: The hands of a clock coincide only once between 11 O' clock and 1 O' clock, so in every 12 hours, the hands of a clock will coincide for 11 times.

$\therefore$ In a day or 24 hours, the hands of a clock will coincide for 22 (11+11) times.

23) How much does a watch lose per day, if its hands coincide every 63 minutes?

A. $56\frac{8}{77}$ min.

B. $57\frac{8}{77}$ min.

C. $55\frac{8}{77}$ min.

D. $54\frac{8}{77}$ min.

The Correct answer is (A)

Explanation: The hands of a clock coincide once after $65\frac{5}{11}$ minutes in every hour.

As per question, the hands coincide in every 63 minutes.

Therefore, the loss in 63 minutes = $65\frac{5}{11} - 63$

$$= \frac{720}{11} - 63$$

$$= \frac{720 - 693}{11} = \frac{27}{11} \text{ minutes}$$

Now, loss in 63 minutes = $\frac{27}{11}$ minutes

$\therefore$ Loss per day or in 24 hours = $\frac{27}{11} * \frac{1}{63} * (24 * 60)$ minutes

$$= \frac{4320}{77} = 56\frac{8}{77} \text{ minutes}$$

24) A man 1.5 meter tall is $30\sqrt{3}$ meter away from a building. When he sees the top of building the angle of elevation is 30° . Find the height of the building?

A. 31.5m

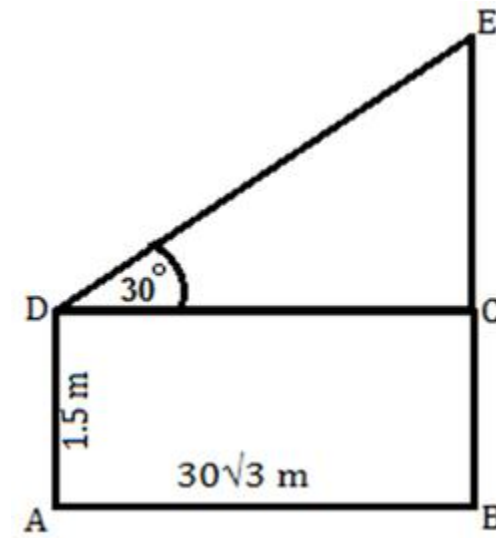
B. 30 m

C. 53.4m

D. 48 m

The Correct option is (A)

Answer with explanation:



Let AD be the man and BE the building.

Now draw a line DC so that DC is perpendicular to BE.

Now, $BC = AD = 1.5$ m

And, $DC = AB = 30\sqrt{3}$ m

$$\frac{EC}{DC} \left(\frac{\text{Perpendicular}}{\text{Base}} \right) = \tan 30^\circ \left(\frac{1}{\sqrt{3}} \right)$$

$$\frac{EC}{30\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$EC = \frac{30\sqrt{3}}{\sqrt{3}} = 30 \text{ m}$$

Now, Height of building, $BE = BC + EC$

$$= 1.5 + 30 = 31.5 \text{ m (Option A)}$$

25) A plane is flying over two villages which are 4 km apart. If at an instant the angles of depression of these villages from the plane are 45° and 60° respectively, find the height of the plane at that instant?

$$1. \frac{4\sqrt{3}}{\sqrt{3}} \text{ km}$$

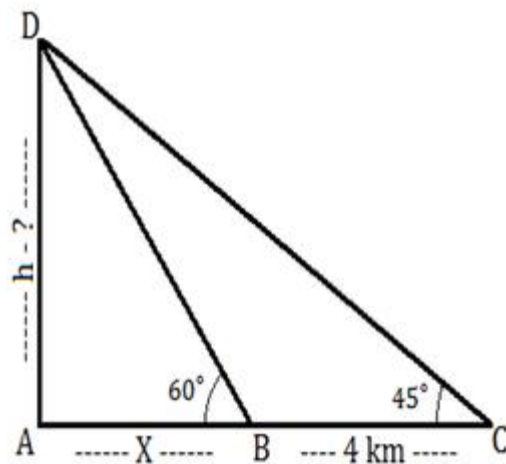
$$2. \frac{2\sqrt{3}}{\sqrt{3}-1} \text{ km}$$

$$3. \frac{2\sqrt{3}}{\sqrt{3}} \text{ km}$$

$$4. \frac{4\sqrt{3}}{\sqrt{3}-1} \text{ km}$$

The Correct option is(4)

Answer with explanation:



Let the plane is at point D and villages at points B and C respectively.

Let the height of the plane be $AD = h$.

$$\text{From } \triangle ABD, \frac{AD}{AB} \left(\frac{\text{Perpendicular}}{\text{Base}} \right) = \tan 60^\circ (\sqrt{3})$$

$$\frac{h}{x} = \sqrt{3}$$

$$x = \frac{h}{\sqrt{3}} \dots\dots\dots(1)$$

$$\text{From } \triangle ACD, \frac{AD}{AC} \left(\frac{\text{Perpendicular}}{\text{Base}} \right) = \tan 45^\circ (1)$$

$$\frac{h}{x+4} = 1$$

$$h = X + 4 \dots\dots\dots(2)$$

Put value of X in equation (2) from equation (1)

$$h = \frac{h}{\sqrt{3}} + 4$$

$$h - \frac{h}{\sqrt{3}} = 4$$

$$h \left(1 - \frac{1}{\sqrt{3}}\right) = 4$$

$$h \left(\frac{\sqrt{3}-1}{\sqrt{3}}\right) = 4$$

$$h = \frac{4\sqrt{3}}{\sqrt{3}-1}$$

$$h = \frac{4\sqrt{3}}{\sqrt{3}-1} \text{ km (Option D)}$$

25) An exam comprises 80 questions carrying one mark each, Parvez answers 65% of the first 40 questions correctly. How much percent of the remaining 40 questions does he need to answer correctly to score 75% in this exam?

- a. 80%
- b. 85%
- c. 73%
- d. 40%

Answer: b

Explanation: Total marks = 80

65% of 40 = $(65/100) * 40 = 26$

And, 75% of 80 = $(75/100) * 80 = 60$

i.e., the required marks $60 - 26 = 34$

So, he needs to answer 34 questions correctly out of the remaining 40 questions.

The required percentage = $34/40 * 100 = 85\%$

26) In an election, 2 candidates participate in which the winner gets 60% of the total votes, and he wins the election by 2000 votes. Find the total number of votes in the election.

- a. 3200
- b. 6400
- c. 10000
- d. 13250

Answer: c

Explanation: Let total votes= 100

So, as per question, the winner gets 60 votes and the loser gets 40 votes.

Now, the difference between votes = 20, but ATQ, it is 2000.

So, we have to multiply 20 by 100 to make it equal to 2000.

So, 100 (the assumed total votes) are also required to be multiplied with 100.

Therefore, the total votes= $100 * 100 = 10000$.

27) Three pipes A, B and C can fill a cistern in 8 minutes, 12 minutes and 16 minutes respectively. What is the time taken by three pipes to fill the cistern when they are opened together?

- A. 3.7 minutes
- B. 4 minutes
- C. 4.5 minutes
- D. 5 minutes

Correct answer; option (A)

Answer with explanation: Part of the tank filled by A in one minute = $\frac{1}{8}$

Part of the tank filled by B in one minute = $\frac{1}{12}$

Part of the tank filled by C in one minute = $\frac{1}{16}$

Net part of the tank filled by A+B+C in one minute;

$$= \frac{1}{8} + \frac{1}{12} + \frac{1}{16}$$

$$= \frac{6 + 4 + 3}{48} = \frac{13}{48}$$

$\frac{13}{48}$ Part of the cistern is filled in one minute.

$\therefore$ The whole tank will be filled in $\frac{48}{13} = 3.7$ minutes

28) A cylindrical tank of diameter 25 cm is full of water. If 11 litres of water is drawn off, the water level in the tank will drop by (use $\pi = 22/7$).

A. $10 \frac{1}{2}$ cm

B. $12 \frac{6}{7}$ cm

C. 14 cm

D. $22 \frac{2}{5}$ cm

The correct answer is D.

Answer with explanation:

Volume of cylinder = $\pi r^2 h$

$$\pi r^2 h = 11 \text{ litres} = 11000 \text{ cm}^3$$

$$\text{Or } \frac{22}{7} * \frac{25}{2} * \frac{25}{2} = 11000 \text{ cm}^3$$

$$h = \frac{11000 * 7 * 2 * 2}{22 * 25 * 25}$$

$$h = \frac{28*4}{5} = \frac{112}{5} = 22 \frac{2}{5} \text{ cm}$$

Therefore, the water level in the will be drop by $222/5$ cm

29) A shopkeeper purchases a table and sells it for Rs. 4200. If he incurs a loss of 20%, find the cost price of table.

- A. Rs.5250
- B. Rs.5150
- C. Rs.5052
- D. Rs.5200

The Correct answer is (A)

Answer with explanation:

S.P. of the table = Rs. 4200

C.P. of the table =?

Loss incurred = 20%

Apply formula: Cost Price = $\frac{100}{100 - \text{loss}\%} \times \text{S.P.}$

$$\text{C.P.} = \frac{100}{100 - 20} \times 4200$$

$$= \frac{100}{80} \times 4200$$

$$= 5250$$

30) Ayaan bought 30 kg of rice at the rate of Rs 9.50/ kg and 40 kg rice at the rate of Rs 8.50/kg and mixed them. He sold the mixture at the rate of Rs. 8.90/kg. Find his total profit or loss in the transaction.

- A. Rs 2 loss
- B. Rs 5 profit
- C. Rs 2 profit
- D. Rs 7 loss

Answer: A

Explanation:

The total cost price = $30 \times 9.5 + 40 \times 8.5 = 625$

The total selling price = $8.9(30+40) = 623$

cost price > selling price = Loss

Hence, the total loss in this transaction = $625 - 623 = \text{Rs } 2$

31) A running man crosses a bridge of length 500 meters in 4 minutes. At what speed he is running?

- A. 8.5 km/hr
- B. 7.5 km/hr
- C. 9.5 km/hr
- D. 6.5 km/hr

The correct answer is (B)

Answer with explanation:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

- o Distance = 500 meters
- o Time = 4 minutes $\rightarrow 4 \times 60 = 240$ seconds

$$\text{Speed} = \frac{500}{240} = \frac{25}{12} \text{ m/s}$$

We need answer in km/hr:

$$\text{Speed in km/hr} = \frac{25}{12} \times \frac{18}{5} = \frac{90}{12} \rightarrow \frac{30}{4} = 7.5 \text{ km/hr}$$

32) Two buses 200 km apart, start moving from the bus stations at the same time towards each other on parallel roads. If they cross each other at a distance of 120 km from one of the bus stations, find the ratio of their speeds.

- A. $\frac{4}{2}$
- B. $\frac{5}{2}$
- C. $\frac{3}{2}$
- D. $\frac{6}{2}$

The Correct answer is (C)

Answer with explanation:

The total distance between the bus stations is 200 km. So, when the buses cross each other the sum of the distances travelled the buses will be equal to 200 km.

We have given that buses cross each other at a distance of 120 km from one of the stations. So one bus covers a distance of 120 km and the other bus covers the rest of the distance ($200 - 120 = 80\text{km}$).

The ratio of their speeds will be equal to the ratio of distances covered by them.

So, ratio of speeds = $\frac{120}{80} \rightarrow \frac{3}{2}$ (Option C)

33) A moneylender charges Rs. 180 as simple interest on a sum of Rs. 600 for four months. What is the rate of interest per annum?

- A. 80%
- B. 85%
- C. 90%
- D. 95%

The Correct answer is (C)

Answer with explanation:

Principal = Rs. 600

Interest = Rs. 180

$$\text{Time} = 4 \text{ months} \rightarrow \frac{4}{12} = \frac{1}{3} \text{ years}$$

Rate of interest = ?

Apply formula: $\text{Rate of Interest} = \frac{I * 100}{P * t}$

$$R = \frac{180 * 100}{600 * \frac{1}{3}}$$

$$= \frac{180 * 100}{200} = 90\%$$

34) A train of length 200 meters crosses a man running at 10 km/hr in the same direction in 10 seconds. What is the speed of the train?

- A. 72 km/hr
- B. 95 km/hr

C. 85 km/hr

D. 82 km/hr

The correct option is (D).

Answer with explanation:

When the train and man are moving in same direction then relative speed will be the difference between their individual speeds. In this problem the other way to find the relative speed is to divide the distance covered (length of train) by the time taken by the train to cross the man.

$$\text{Relative Speed} = \frac{200}{10}$$

We will convert it into Km/hr

$$\frac{200}{10} \times \frac{18}{5} = 72 \text{ km/hr}$$

Now, let the speed of the train is X km/hr. So, the relative speed, 72 km/hr = X km/hr - 10 km/hr

$$X - 10 = 72$$

$$X = 72 + 10$$

$$X = 82 \text{ km/hr}$$

35) A can finish a work in 8 hours, B can finish the same work in 12 hours, and C can do it in 24 hours. All the three starts work at 3 am, 4 am, and 5 am respectively. At what time work will be completed?

A. 6 am

B. 7: 40 am

C. 8 am

D. 5 am

Answer: B

Explanation:

Note: Assume the total work = LCM of the given time

Take the LCM of hours = LCM of (8, 12, and 24) = 24

Let the total work = 24

Note: One hour work = (total work/ time)

Now,

A's one hour work = $24/8 = 3$ unit

B's one hour work = $24/12 = 2$ unit

C's one hour work = $24/24 = 1$ unit

ATQ, between 3 am - 4 am, A works alone and finish 3 units of work

Between 4 am - 5 am, A and B work and complete $3+2 = 5$ units of work

Between 5 am - 6 am, A, B, and C works and finishes $3+2+1 = 6$ units of work

Or, between 3 am to 5 am, 8 unit of work is finished.

Remaining $24-8 = 16$ unit (till 5am)

After 5 am, A+B+C work together and finish 6 unit of work per hours.

i.e., 6 unit work = 1 hour

To complete the 16 unit work, multiply both sides with 2

i.e., 12 unit work = 2 hours

Remaining work = $16-12 = 4$ unit work

As we know, 6 work = 1 hour

Or, 1 work = $1/6$ hours

Or, 4 work = $4/6$ or $2/3$ hours.

That means $(2 + 2/3)$ hours requires after 5 am.

So, time taken to complete the work is $(5 + 2 + 2/3) = 7:40\text{am}$

36) A 20 liter mixture contains 30% alcohol and 70% water. If 5 liters of water is added to the mixture, what will be the percentage of alcohol in the new mixture ?

A. 22%

B. 23%

C. 24%

D. 25%

Hide Answer Workspace

The Correct answer is (C)

Answer with explanation:

Initially, the mixture contains 30% alcohol = $\frac{30}{100} * 20 = 6$ liters of alcohol

And, 70% of water = $\frac{70}{100} * 20 = 14$ liters of water

After adding 5 liters of water, the mixture contains (14+5) 19 liters of water and 6 liters of alcohol.

∴ Percentage of alcohol = $\frac{6}{25} * 100 = 24\%$

37) Value of $(256)^{0.16} \times (256)^{0.09}$ is

- A. 64
- B. 4
- C. 8
- D. 16

Answer: B

Explanation:

$$\begin{aligned} & (256)^{0.16} \times (256)^{0.09} \\ & (256)^{0.16+0.09} = (256)^{0.25} = (256)^{1/4} \\ & = (4^4)^{1/4} = 4^{4 \times 1/4} = 4 \end{aligned}$$

38) Which least number should be subtracted from 1936 so that the resulting number, when divided by 9, 10, and 15, will leave the same remainder 8 in each case?

- A. 39
- B. 38
- C. 37
- D. 36

Answer: B

Explanation:

Take LCM of 9, 10, and 15. Their LCM is = 90

If the number 1936 is divided by 90, we get the remainder 46, but we need 8 as remainder.

Hence, the required number is $46 - 8 = 38$.

39) The sides of a triangle are in the ratio 10: 24:26 and its perimeter is 300 m. What is its area?

A. 2500 m^2

B. 3000 m^2

C. 3500 m^2

D. 4000 m^2

Explanation:

Let the sides are $10x$, $24x$, and $26x$.

The perimeter is 300 m.

$$\text{So, } 10x + 24x + 26x = 300$$

$$60x = 300$$

$$x = 5$$

So, the sides are $10 * 5 = 50$ meters

$$24 * 5 = 100 \text{ meters}$$

$$26 * 5 = 130 \text{ meters}$$

$10^2 + 24^2 = 26^2$ so, it is a right triangle.

$$100 + 576 = 676$$

The area of a right triangle is $= \frac{1}{2} * \text{base} * \text{height}$

$$= \frac{1}{2} * 50 * 100$$

$$= 2500 \text{ m}^2$$

40) 5600 is to be divided into A, B, C, and D in such a way that the ratio of share of A: B is 1: 2, B: C is 3: 1, and C: D is 2: 3. Find the sum of (A and C) and (B and C).

A. Rs 2400, Rs 3000

B. Rs 2000, Rs 3000

C. Rs 2400, Rs 3200

D. Rs 2000, Rs 3200

Answer: D

Explanation:

ATQ,

A: B = 1: 2..... (i)

B: C = 3: 1..... (ii)

C: D = 2: 3.....(iii)

Now,

Find A: B: C: D

Step 1: A: B: C: D

1: 2 (A: B value by equation i)

Note: To understand the shortcut, remember you need to make the right-hand side missing numbers the same as that of last given number, and for the right-hand side the same is done.

i.e., C: D will contain 2: 2 because 2 is the last number on the right side.

Or, A: B: C: D

1: 2: 2: 2

3: 3: 1: 1 (B: C value by equation ii)

2: 2: 2: 3 (C: D value by equation iii)

Now, multiply vertically and to get A: B: C: D.

So, A: B: C: D = $(1 \times 3 \times 2): (2 \times 3 \times 2): (2 \times 1 \times 2): (2 \times 1 \times 3)$

= 6: 12: 4: 6

= 3: 6: 2: 3

Now, the share of A and C = $[(A+C)/ (A+B+C+D)] \times \text{total amount}$

Or, the share of A and C = $[(3+2)/ (3+6+2+3)] \times 5600$

Or the share of A and C = $(5/14) \times 5600 = 2000$

Similarly, the share of B and C = $(8/14) \times 5600 = 3200$

41) In a race of 600 m, A can beat B by 50 m, and in another race of 500 m, B can beat C by 60 m. Then by how many meters will A beat C in a race of 400 meters?

A. $77 \frac{1}{3}$ m

B. 77 m

C. 70 m

D. None of these.

Answer: A

Explanation:

When A runs 600 m,

Then B runs $(600 - 50) = 550$ m

Now, When B runs 500 m,

Then C runs $(500 - 60) = 440$ m

When B runs 550 m then C runs = $\frac{440 \times 550}{500} = 484$ m

When A runs 600 m then C runs 484 m,

Now in a 400 meters race, when A runs 400 runs, C will run: $\frac{484}{600} \times 400 = \frac{968}{3}$ m

$\therefore$ A beats C in a 400 m race by $\left(400 - \frac{968}{3}\right) = \frac{1200 - 968}{3} = \frac{232}{3} = 77\frac{1}{3}$ m

42) A box contains 2 red, 3 green, and 2 blue balls. What is the probability that none of the balls drawn is blue?

- A. 10/25
- B. 10/21
- C. 10/31
- D. 10/21

Answer: B

Explanation:

Total number of balls = $(2+3+2) = 7$

Let S be the sample space.

Then, $n(S)$ = the total number of ways of drawing two balls out of 7:

$$= {}^7C_2 \\ = \frac{7 \times 6}{2 \times 1} = 21$$

Let E is the event of drawing 2 balls, none of which is blue.

$n(E)$ = number of ways of drawing 2 balls out of $(2+3)$ balls.

$$= {}^5C_2$$

$$= \frac{5 \times 4}{2 \times 1} = 10$$

$$\text{So, } P(E) = \frac{n(E)}{n(S)} = \frac{10}{21}$$

43) In a group of students, there are 15 boys and 10 girls. If three students are selected at random, what is the probability that 1 girl and 2 boys are selected?

- A. 21/46
- B. 21/36
- C. 21/26
- D. 21/56

Answer: A

Explanation:

Let sample space = S.

n(S) = number of ways to select 3 students (1 girl and 2 boys) out of 25 students

$$= {}^{25}C_3$$

$$= \frac{25 \times 24 \times 23}{3 \times 2 \times 1}$$

$$= 2300$$

$$n(E) = {}^{10}C_1 * {}^{15}C_2$$

$$= 10 * \frac{15 \times 14}{2 \times 1}$$

$$= 1050$$

Apply formula:

$$P(E) = \frac{n(E)}{n(S)} = \frac{1050}{2300} = \frac{21}{46}$$

44) 4 cubes each of 6 cm² total surface area and another cube of 24 cm² total surface area are placed together in such a way to form a rectangular parallelepiped. The total surface area of the parallelepiped so formed is

- A. 48 cm²
- B. 40 cm²

- C. 36 cm^2
- D. 32 cm^2

Answer: A

Explanation: Total surface area of parallelepiped = total surface area of 4 cubes + Surface area of another cube

So, the required surface area = $4 \times 6 + 24 = 48 \text{ cm}^2$

45) What is the value of $(25.732)^2 - (15.732)^2$?

- A. 414.64
- B. 414.256
- C. 414.128
- D. 414.52

Answer: A

Explanation: We know that, $a^2 - b^2 = (a+b)(a-b)$

Or, $(25.732)^2 - (15.732)^2 = (25.732 + 15.732)(25.732 - 15.732)$

Or, the value = $41.464 \times 10 = 414.64$

46) If $1.125 \times 10^k = 0.001125$, the value of k is

- A. -4
- B. -5
- C. -3
- D. -1

Answer: C

Explanation:

The given expression is $1.125 \times 10^k = 0.001125$

Or, we can say $0.001125 = 1.125/10^3 = 1.125 \times 10^{-3}$

Now, $1.125 \times 10^k = 1.125 \times 10^{-3}$

On comparing both sides, we get $10^k = 10^{-3}$

Hence, $k = -3$